KULBHUSHAN SINGH

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SUMMARY

Performance-driven C++ Video Processing Engineer with 3+ years of experience designing, optimizing, and deploying real-time video pipelines. Expert in video codecs (H.264/AV1), MPEG2 videos. Proven record of improving system performance by 70%+, with deep knowledge in CNNs, motion estimation, and signal processing. Passionate about building scalable, intelligent systems with practical impact.

SKILLS

C, Modern C++ (17/20), STL, Multithreading, Async I/O, Python, Pytorch, Git/Github, Image Processing, Video Processing, Computer Vision, Object Detection, Signal Processing, Optimization, FFMPEG, Applied Mathematics, Debugging, H. 264 Codec Expert

WORK EXPERIENCE

Software Engineer (Video Quality), Interra Systems — **Jan 2023 - Present**

- Contributions to no-reference video quality assessment algorithms for QC automation in Baton Product.
- Replaced block-based motion estimation with a multi-scale pipeline using image pyramids and optical flow (Pyramidal Lucas–Kanade, Horn–Schunck, Farneback), fused by a Kalman filter and RANSAC to robustly and accurately recover rotation, translation, and zoom.
- 40% Optimization in legacy motion vector estimation with near 0% accuracy trade off.
- Implemented Fast Zero Normalized cross correlation, surpassing the best benchmark for accuracy.
- Implemented a production-grade scene-change detector achieving 30% runtime reduction and 97% precision / 100% recall under flash, high-motion, and low-light conditions.
- Integrated 16-bit depth support in black bar detection for UHD and 4K videos, improving detection by 40%.
- Researched and implemented Out-of-Gamut pixel detection check for various RGB Color Space Videos.
- Optimized Chroma Dropout, CIE Color Gamut, and Blank Bars using Intel intrinsic, achieving a 71% performance gain in Black Bars and 40% in Chroma Dropout.
- Researched and developed new algorithms for Blocky Dropout and Concealment Artifacts in Videos, surpassing legacy Accuracy by 50% and performance gain up to 80%.
- Introduced Git version control and planned CI/CD integration for the Video Quality team for smooth release.
- Experience in Applied Mathematics for video quality assessment.
- Developed a new feature to find No reference Video quality index using Contrastive learning for H.264 encoded streams.

Subject Matter Expert Intern, Paathshala Education — **Nov 2021 - May 2022**

- Taught Advanced Mathematics concepts for IIT-JEE Mains & Advanced, covering Calculus, Algebra, Coordinate Geometry, Trigonometry, and Probability.

OPEN SOURCE CONTRIBUTIONS — Jun 2025 - Present

- **TorchMetrics (PyTorch Ecosystem) – Ongoing contributor**
  - Implemented reduction='none' support for the VIF metric, enabling per-sample evaluation.
  - Fixed regression issues in classification metrics—e.g., handling edge cases like zero true positives/false positives by setting precision to NaN.
  - Collaborated with maintainers through GitHub pull requests and CI-driven reviews, improving test reliability and PR throughput.

EDUCATION

Master of Computer Application (Software Engineering) — **Oct 2021 - July 2023**
University School of Information, Communication and Technology — CGPA 9.0/10

Bachelor of Computer Applications — **Aug 2018 - July 2021**
Guru Gobind Singh Indraprastha University (GGSIPU) — CGPA 9.06/10

PROJECTS

- **Twitter Sentimental Analysis** **Jan 2023 - July 2023**
  - Developed a sentiment analysis system using the Twitter API to retrieve real-time user tweets.
  - Preprocessed text data using regex for noise removal and structured it with Pandas DataFrame.
  - Applied Text Polarity and Sentiment Mining techniques to classify tweet sentiments.
  - Displayed processed sentiment data on the front end for real-time analysis.

ADDITIONAL INFORMATION

- **Languages:** English, Hindi, German
- **Certifications:** Data Structures and Algorithms Nano Degree by Udacity
- **Achievements:** AIR 1100 in NIMCET Exam
- **Patent:** Filed a Patent for detection of Video Dropout detection **(Ist Inventor)**